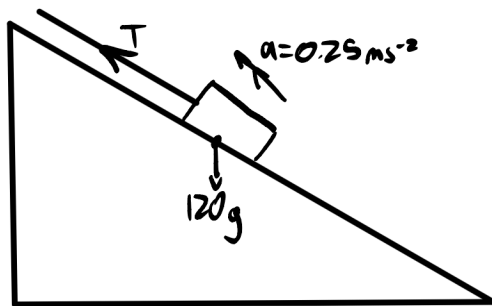


Consolidation

1. i.



$$\begin{aligned} \text{ii.} \quad T - 120g \sin(10) &= 120 \times 0.25 \\ T &= 234 \text{ N} \end{aligned}$$

$$2. \quad \text{i.} \quad 55g \sin(20) - 30 = 55a \quad a = 2.81$$

$$\begin{aligned} \text{ii.} \quad v &= u + at \\ &= 0 + 2.8 \times 5 \\ &= 14 \text{ ms}^{-1} \end{aligned}$$

$$\begin{aligned} 3. \quad 5g \sin(\theta) &= 5a \\ a &= 1.96 \end{aligned}$$

$$\begin{aligned} s &= \frac{v^2 - u^2}{2a} \\ &= \frac{7^2}{2(1.96)} = 12.5 \text{ m} \end{aligned}$$

$$\begin{aligned} 4. \quad \text{i.} \quad 30g \sin \alpha - 50 &= 30a \\ a &= \frac{\frac{150 \times 9.8}{13} - 50}{30} = 2.10 \end{aligned}$$

$$\begin{aligned} \text{ii.} \quad v &= \sqrt{u^2 + 2as} \\ &= \sqrt{0 + 2 \times 2.103 \times 5} \\ &= 4.59 \text{ ms}^{-1} \end{aligned}$$

$$\begin{aligned} \text{iii.} \quad -50 &= 30a \\ a &= -1.67 \end{aligned}$$

So deceleration is 1.67 ms^{-2} .

$$\begin{aligned} \text{iv.} \quad v &= \sqrt{u^2 + 2as} \\ &= \sqrt{4.59^2 + 2 \times -1.67 \times 5} \\ &= 2.09 \text{ ms}^{-1} \end{aligned}$$

5.

$$R = 20 \cos 30$$

$$20 \sin 30 - \mu R = 0$$

$$20 \sin 30 - \mu(20 \cos 30) = 0$$

$$\mu = \tan 30$$

$$= 0.577$$

6. i.

$$R = 20g \cos \left(\arcsin \left(\frac{7}{25} \right) \right)$$

$$F_{\max} = \mu R = 4g \cos \left(\arcsin \left(\frac{7}{25} \right) \right)$$

To stop the block sliding down the plane, the force F acting parallel to the plane must be equal to the maximum friction force $F_{\max}$ minus the weight acting down the plane $\frac{7}{25}W$. Then

$$F_{\max} = 37.632$$

$$\begin{aligned} \therefore F &= \left(\frac{7}{25} \times 20 \times 9.8 \right) - F_{\max} \\ &= 17.2\text{N} \end{aligned}$$

ii. In this scenario, the force F required to move the block up the plane must oppose both the friction force $F_{\max}$ and the weight of the block acting down the plane $\frac{7}{25}W$. Then

$$F_{\max} = 37.632$$

$$\begin{aligned} \therefore F &= \left(\frac{7}{25} \times 20 \times 9.8 \right) + F_{\max} \\ &= 92.5\text{N} \end{aligned}$$

7.

$$R = 50g \cos 30 - T \sin 20$$

$$F_{\max} = \mu R = \frac{1}{4}(50g \cos 30 - T \sin 20)$$

i. If the mass is about to move down the slope, $T \cos 20 + F_{\max} = 50 \sin 30$.

$$T \cos 20 + \frac{1}{4}(50g \cos 30 - T \sin 20) = 50g \sin 30$$

$$4T \cos 20 + 50g \cos 30 - T \sin 20 = 200g \sin 30$$

$$T(4 \cos 20 - \sin 20) + 50g \cos 30 = 200g \sin 30$$

$$\begin{aligned} T &= \frac{200g \sin 30 - 50g \cos 30}{4 \cos 20 - \sin 20} \\ &= 162.6\text{N} \end{aligned}$$

ii. If the mass is accelerating at 5ms^{-2} up the slope, $T \cos 20 - F_{\max} - 50 \sin 30 = 250$

$$T \cos 20 - \frac{1}{4}(50g \cos 30 - T \sin 20) - 50 \sin 30 = 250$$

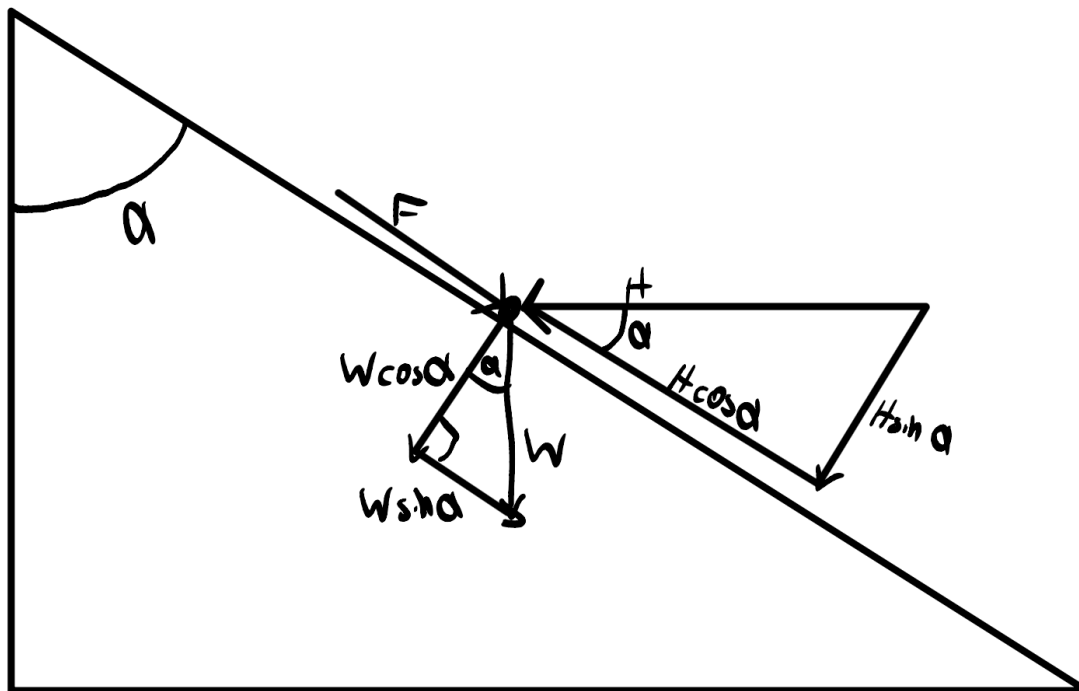
$$4T \cos 20 - 50g \cos 30 + T \sin 20 - 200 \sin 30 = 1000$$

$$T(4 \cos 20 + \sin 20) - 50g \cos 30 - 200 \sin 30 = 1000$$

$$\begin{aligned} T &= \frac{1000 + 50g \cos 30 + 200 \sin 30}{4 \cos 20 + \sin 20} \\ &= 372\text{N} \end{aligned}$$

Exam Questions

1. a.



b.

$$\alpha = \arctan\left(\frac{4}{3}\right) = 53.1301^\circ$$

$$R = W \cos \alpha + H \sin \alpha$$

$$F_{\max} = \mu R = \frac{1}{2}(W \cos \alpha + H \sin \alpha)$$

As H is the greatest value for which P remains in equilibrium, then

$$H \cos \alpha - W \sin \alpha = \frac{1}{2}(W \cos \alpha + H \sin \alpha)$$

$$2H \cos \alpha - 2W \sin \alpha = W \cos \alpha + H \sin \alpha$$

$$H(2 \cos \alpha - \sin \alpha) = W(\cos \alpha + 2 \sin \alpha)$$

$$\begin{aligned} H &= \frac{\cos \alpha + 2 \sin \alpha}{2 \cos \alpha - \sin \alpha} W \\ &= \frac{11}{2} W \end{aligned}$$

c.

$$mg \sin \alpha - \frac{1}{2}(mg \cos \alpha) = ma$$

$$g \sin \alpha - \frac{1}{2}g \cos \alpha = a$$

$$g\left(\sin \alpha - \frac{1}{2} \cos \alpha\right) = a$$

$$\frac{1}{2}g = a$$

Therefore, $a = \frac{1}{2}g$